

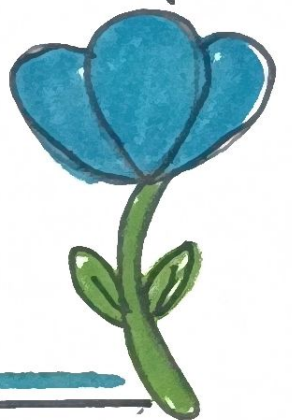


ASSESSMENT

TOOI-3

ITA1412 - Ethical
Hacking

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1. A student used Nmap to scan a company's network without permission. No data was stolen, but the company reported the activity. Is the student's action ethical and legal?

Ethical Analysis:

- The student's action is not ethical because permission was not taken
- Ethical hacking always requires authorization
- Good intention does not make unauthorized scanning acceptable.

Legal Analysis:

under the IT Act, 2000:

- section 43: unauthorized access to computer systems
- section 66: Computer-related offences.

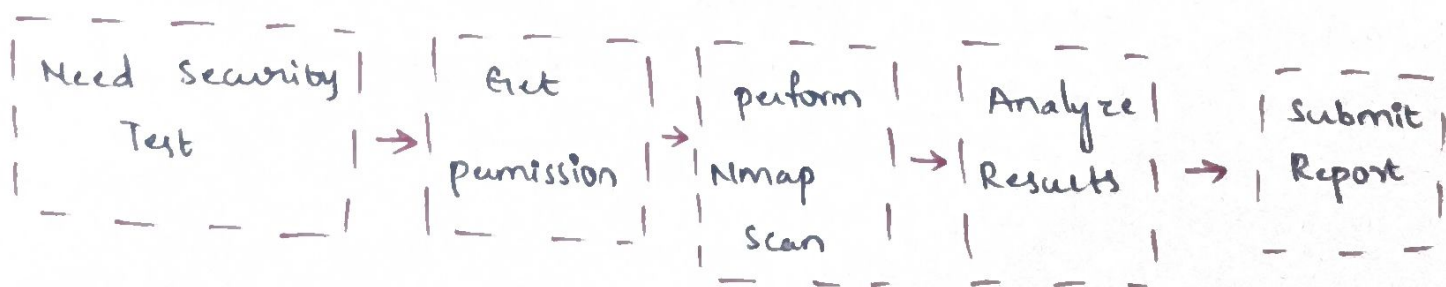
Consequences:

- Cybercrime investigation
- Fine or legal action
- Damage to reputation

Correct procedure:

- Take written permission
- Define the testing scope
- perform the scan.
- Submit the security report.

Flow chart:



Conclusions

Ethical hacking always requires permission from the system owner before testing. Scanning a network without authorization is unethical and may violate the IT Act, 2000. The student should have obtained written permission before using NMap.

2. A researcher found a SQL injection vulnerability and posted it online before informing the company. Hackers used it to steal customer data. Was this correct?

Ethical Analysis:

- The researcher acted unethically
- The company should have been informed privately first

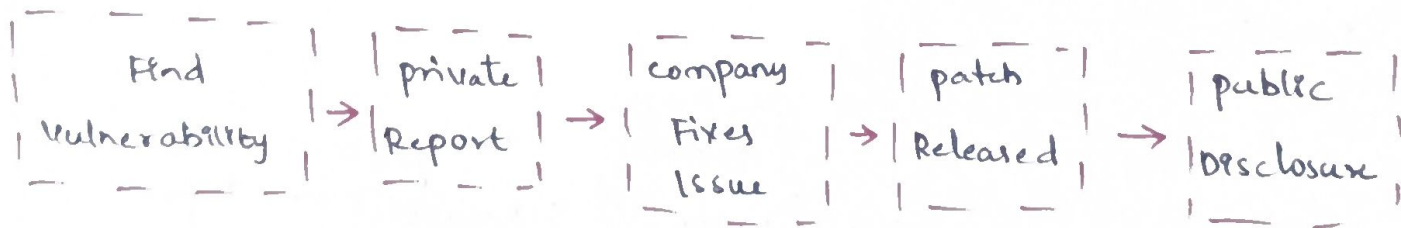
Responsible Disclosure:

- Find the vulnerability
- Report it privately
- Wait for the fix
- publish details later.

Risks:

- Customer data theft
- Financial loss.
- Reputation cyberattacks
- More cyberattacks.

Flow Chart:



Conclusion:

Security vulnerabilities should be reported privately to the organization first. public disclosure before a fix can lead to cyber attacks and data theft. Responsible disclosure helps protect users, organizations, and sensitive information.

3. An employee received a fake email asking for login details and entered the password. The attacker accessed company data. Identify the attack and explain prevention.

Attack Type:

phishing attack

IT Act Sections:

- Section 43 : unauthorized access
- Section 66 : computer offenses.
- Section 66c : Identity theft
- Section 66D : online cheating

Responsibilities:

Organization

- Employee training
- Enable MFA
- Secure email filtering

Employee

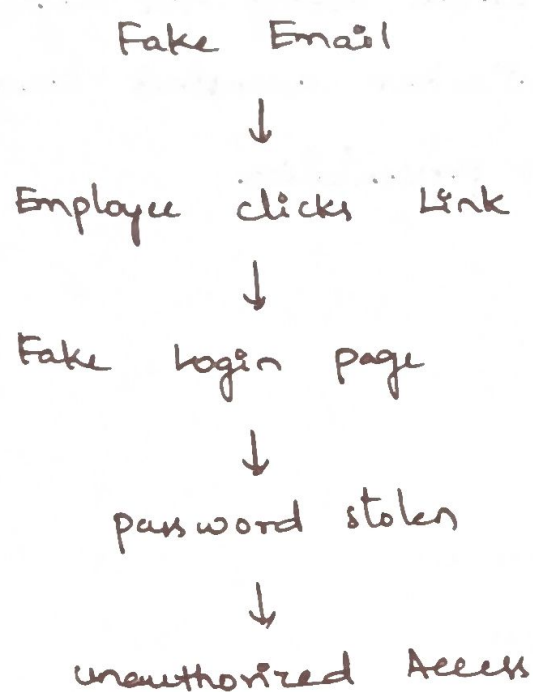
- Verify sender
- Avoid suspicious links.
- Report phishing emails

Prevention:

- Multi-Factor Authentication

- Email Filtering
- Security awareness training
- Strong password policy.

Flowchart:



Conclusion:

phishing attacks target people by using fake emails and websites to steal information. Organizations and employees must work together to prevent such attacks. Training, awareness, and strong security measures can greatly reduce phishing risks.

4. A student found a banking website vulnerability and accessed customer information before reporting it. Was the action ethical and legal?

Ethical Analysis:

- The action is not ethical
- Customer data should never be accessed without

permission

Ethical Vs unauthorized Hacking

Ethical Hacking	unauthorized Hacking
Permission taken	No permission
Legal	Illegal
protects systems	Misuses systems

IT Act Sections:

- section 43: unauthorized access
- section 66: computer-related offences.

Correct Reporting:

- Confirm the issue
- collect evidence
- Report privately
- keep information confidential

Flowchart:



conclusions:

Good intentions do not make unauthorized access legal or ethical. Ethical hackers must always obtain permission before testing any system. Following responsible reporting procedures help protect users and avoid legal consequences.